DATA CENTER FACILITIES

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Challenges for High Density Workloads in Modular Data Centers

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High density changes the modular design basis

- Compute rack power rises faster than room and distribution assumptions.
- Direct-to-chip cooling shifts heat to liquid but leaves residual air loads.
- Power, fluid, data, structure and safety converge at each rack position.
- The module must absorb changing equipment mixes without losing service access.

Seven challenges shape the deployment

- Uneven rack density and fast equipment refresh cycles
- Power distribution and fault coordination at the rack
- Liquid delivery, residual air and transient heat
- Space, weight, transport and maintenance access
- Controls, fault response and life safety
- Vendor interfaces and repeatable acceptance tests

A mixed rack population is the starting point

- Compute, fabric, storage and support racks have different duty cycles.
- Some racks remain air cooled while others rely on direct-to-chip liquid.
- A common room setpoint does not describe every rack inlet condition.
- Each rack schedule must state peak power, liquid duty and residual air duty.

The rack row is a shared infrastructure boundary



Illustrative modular containment and rack service architecture.

Power delivery has to follow the rack schedule

- Map A and B feeds to the actual OEM power connection count.
- Select busway taps, feeders and protection for each rack duty.
- Verify short circuit ratings, selectivity, grounding and emergency power off.
- Reserve accessible routing and replacement space above the row.

The overhead platform organizes services



Power, liquid and pathway interfaces can share a project-engineered service structure.

Liquid cooling requires a complete hydraulic model

- Start with OEM heat capture, supply temperature, flow and pressure limits.
- Model mains, rack branches, valves, quick disconnects and CDU duty together.
- Define isolation so maintenance or a leak has a controlled service boundary.
- Confirm coolant chemistry, water quality and commissioning procedures.

Liquid distribution reaches every rack branch



Headers, branch connections and isolation need clear service access.

Residual air cooling remains essential

- Memory, network equipment, power supplies and rack electronics still heat air.
- Contain return air and measure conditions at the rack inlet.
- Account for fault and transient conditions when liquid duty changes.
- Adjust airflow as the rack population converts from air to hybrid operation.

The modular envelope creates physical tradeoffs

- Service clearances compete with rack count and transport dimensions.
- Busway, liquid headers and cable trays require independent support.
- Weight, floor loading, anchoring and seismic criteria affect the layout.
- Door swing, equipment replacement and egress need full-scale checks.

Service access is part of the architecture



A contained hot aisle still needs practical operator and equipment access.

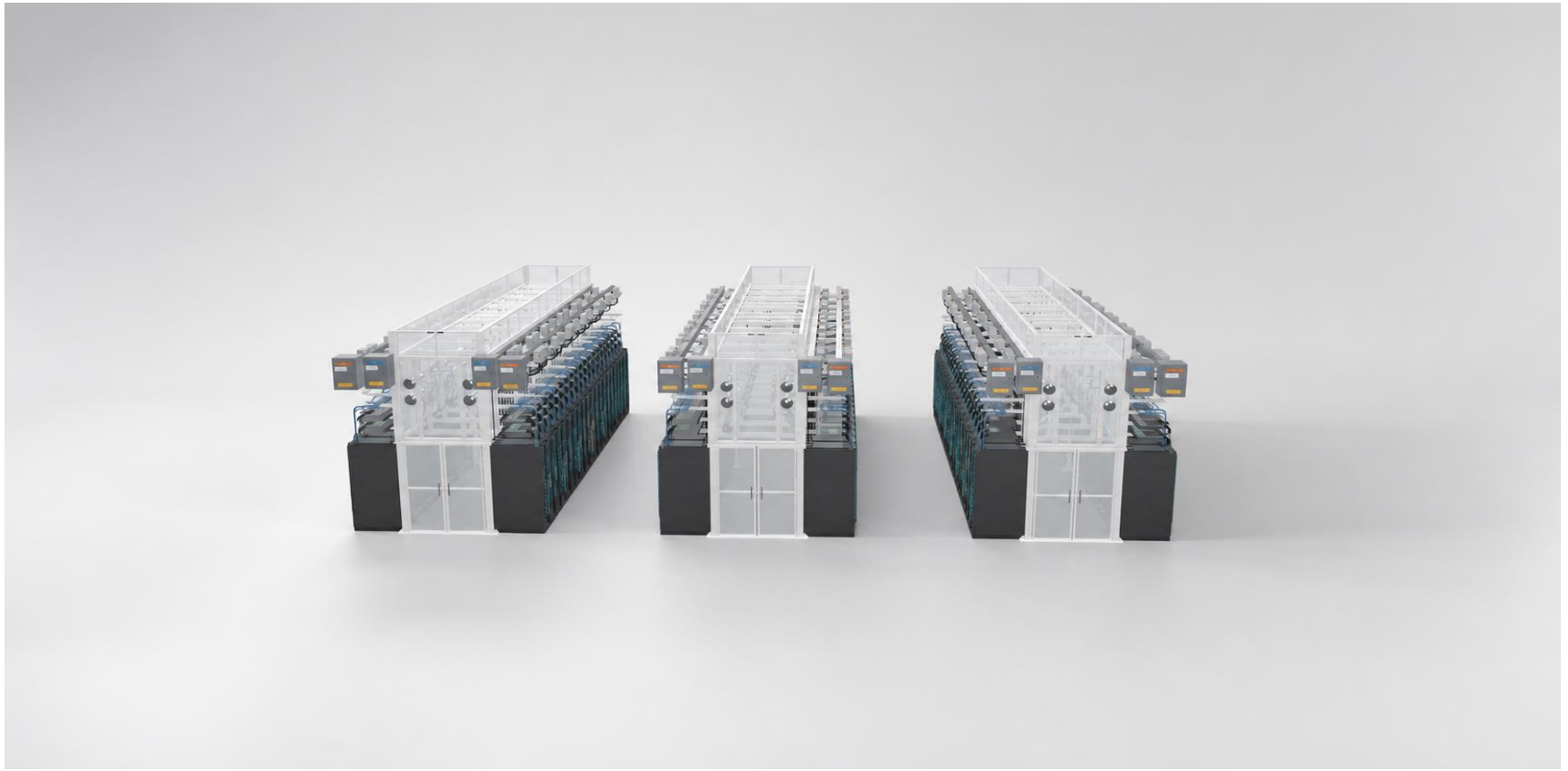
Local controls must handle faults without delay

- Monitor liquid events, valve feedback, temperature and differential pressure.
- Define the sequence for leak isolation and loss of flow or power.
- Coordinate detection, suppression, EPO and door release with the site design.
- Expose consistent alarms and telemetry to BMS or DCIM.

Open interfaces enable mixed vendor deployment

- Publish rack dimensions, connections and tolerance bands.
- Specify power and liquid handoff points by rack position.
- Use common sensor locations, point names and data quality states.
- Make expansion joints and service boundaries explicit.

A modular cell supports phased installation



Cell and pod layouts can add rack capacity in defined stages.

Factory tests should reproduce operating cases

- Inspect assembled structure, connections, labeling and leak integrity.
- Exercise air and liquid operating modes at representative load.
- Prove alarms, sensor loss, valve feedback and emergency sequences.
- Carry test records and open items into site acceptance.

Acceptance criteria need to travel with the module

- Define rack inlet limits and liquid conditions for each operating case.
- Set pressure and leakage criteria for the containment boundary.
- Document power protection and controls test results by configuration.
- Repeat key tests after site connections and expansion.

An OCP interface specification can reduce rework

- Standardize rack, containment, power, liquid and controls terminology.
- Publish dimensional and service access requirements.
- Describe fault scenarios and minimum FAT/SAT evidence.
- Demonstrate interoperability with more than one equipment vendor.

Call to action: define the open HAC boundary

Problem to solve

Air and liquid cooling are converging faster than the interfaces connecting racks, containment, cooling, controls, and life-safety systems.

How to get involved

- Join the OCP Modular Data Center subgroup monthly calls.
- Contribute dimensional requirements, operating modes, failure scenarios, and test cases.
- Volunteer rack, containment, cooling, and controls equipment for interoperability demonstrations.
  - Help define the vendor-neutral MDC-AI HAC reference specification.

Thank You

